

Course Code: SED 102

Course Title: Foundations of Science

Credit Units: 2

What is science?

Definitions of Science vary from scientist to scientist. As early as the 1940's and 1950's it was regarded as a body of classified organized and systematized knowledge. Some defined science as "a group of exact demonstrable facts and proven theories".

Science is a systematic evidence-based enterprise that builds and organizes knowledge. However, when defined in terms of its **contents or products**; science is an ordered body of knowledge in form of facts, principles, laws, theories and concepts about how the universe operates. Also, when defined in terms of its **process**, science is a method of investigating about the universe through objective observation and experimentation.

Science has also been defined as what scientists do? What do scientists do?

Scientists ask questions 'what' and how? But not 'why'? This is because 'why' a question of theology and metaphysics.

Science is a broadly based human enterprise that may be viewed differently by different people. For instance, some scientists view science as a method by which hypotheses are tested. Philosophers view science as a way of questioning the usefulness of knowledge. A lay man may define science as a body of scientific information. According to Collette and Chiapeta (1984), Science should be viewed as a body of knowledge, a way of investigation or method and a way of thinking in the pursuit of an understanding of nature.

Characteristics of scientific knowledge

- 1. Testability:** theories and explanations must be tested and verified or proven false.
- 2. Systematic methodology:** it relies on careful measurement, critical analysis, and reputable experiments.
- 3. Objectivity and Replication:** findings are shared openly so that researchers can independently test, replicate, or challenge the results.

Science is broadly divided into different field based on what is being studied

- Natural science: is the study of the physical world and nature. This includes physics(Matter and energy), Chemistry (atoms and substances), and Biology (life and living organisms)
- Earth and space sciences: is the study of planets, stars, and Earth's system, including fields like geology, meteorology and astronomy.
- Social and formal sciences: the study of human behaviour, societies, and abstract systems like logic and mathematics.

Basic concepts in the teaching and learning of science

- Facts
- Concepts
- Hypothesis
- Theories
- Principles and Laws

Facts

A scientific fact is a truth and the actuality of the statement of things the way they are. A fact does not change and facts are often indisputable. A fact is actually something which is confirmed to a very high degree such that it may not make sense for anybody not to accept it as a fact. Two criteria can be used in identifying a fact;

1. It is directly observable and
2. It can be demonstrated anytime

Concepts

A concept is an aggregation of facts that have shown certain significant relationships and patterns. A concept serves as springboard for further investigations. A general idea derived or inferred from specific instances or occurrences. Something formed in the mind; a thought or notion eg. the Mole Concept is widely applied in Chemistry because it was arrived at from a combination of several facts.

Hypothesis

Hypotheses are scientific guesses and speculations. They could also be referred to as scientific assertions that have not been confirmed to be true. Hypotheses are formulated without backing them up with any empirical evidence.

Scientific Theories

When hypotheses have significantly more empirical supports and proofs, it becomes a theory. Therefore, a theory could be referred to as a scientific speculation that has been empirically supported. It fits all facts and has no reasonable rival. Theories are inferred explanations of some aspect of the natural world. Theories do not become laws even with additional evidence; they explain laws. However, not all scientific laws have accompanying explanatory theories.

Principles and Laws

When a concept is based on so many facts that are all consistent within the limit of observational error, it becomes a scientific principle or law. However, there could be some exceptions to laws, but such exceptions are often very insignificant. Laws are generalizations or universal relationships related to the way that some aspect of the natural world behaves under certain conditions. Well-established laws and theories must be internally consistent and compatible with the best available evidence; be successfully tested against a wide range of applicable phenomena and evidence; possess appropriately broad and demonstrable effectiveness in further research.

Scientific Processes

The Science Teachers Association of Nigeria (STAN) (1988) presented a list of scientific processes as presented by the American Association for the Advancement of Science. These are:

- Experimentation
- Observation
- Interpretation of data
- Classification
- Recording
- Communication
- Manipulation of Variables
- Formulation of Hypotheses
- Making Conclusions/Inference
- Making Prediction
- Measuring
- Making operational definition
- Designing/formulation of models etc.
- Using numbers
- Using space/time relationships
- Questioning

Explanation of some of the processes

Observation: To observe means to look at an occurrence, take note of what is occurring and then make a record of what is happening and think about it deeply to determine the cause of the occurrence and a likelihood of any further event as a result of that which is occurring.

Experimentation: Involves the manipulation of environmental variables like temperature, pressure, humidity, concentration of substances etc. within a set up to verify an already established scientific knowledge or to discover new scientific knowledge. In conducting experiments, scientists play around with matter to observe possible changes and transformations under certain conditions.

Formulation of Hypotheses and Theories: A hypothesis is a reasonable scientific guess which has not been confirmed. It can also be referred to as a scientific speculation. However, if a hypothesis is now supported with significant empirical proofs, it can now advance to the status of a theory. A theory is therefore, a guess or speculation that has been established severally through empirical efforts and conclusions.

Making Conclusions and Inferences: When scientists conduct experiments, they record their observations and subject this observation to rigorous thinking which can make them to arrive at some statements of facts which are called **conclusions**.

Classification: Classification means sorting things out based on similarities and differences.

Communication: We communicate in science to share ideas with fellow scientists about new discoveries in science. The communication can take various forms e.g. by E-mails, publications in journals or verbal communication. It is important to communicate scientific findings so as not to render them useless.

Record: Keeping good account of scientific findings is very important because further discoveries depend on existing records. Scientific records are very useful and therefore must be properly kept so that references can be made to them when needed.

Manipulation of Variables: What are variables? Variables are factors that have the tendency to change the situation or state of an environment or a substance. Examples of variables in science are temperature, pressure, humidity, salinity, concentration, pH etc. because the state of matter can change when these variables change. Scientists always have cause to vary these variables and study their effects on matter.

Introduction to scientific literacy

Scientific Literacy

Scientific literacy could best be defined in term of what a scientifically literate individual should be able to do. This can be expressed in so many ways. For instance,

1. Substantial science background knowledge of the facts, concepts and theories of science and the ability to apply them.
2. An appreciation of the value of science and technology in the society and how Science Technology and Society (STC) influence one another.
3. The ability to use the processes of science to solve problems and to make daily decisions.
4. Possession of science process skill that permits an individual to function in the work place, leisure and in the society in general.

What is science Education?

Science education is the structured teaching and learning of scientific knowledge, inquiry skills, and critical thinking across disciplines like biology, chemistry, physics, and earth sciences. It goes beyond memorizing facts, focusing on how to ask questions, gather evidence, and apply scientific reasoning in everyday life. It can also be defined as the field dedicated to teaching and learning about the natural world, including physical, life, earth, and

space sciences, to cover both content knowledge (scientific concepts) and process skills (experimentation, data analysis, reasoning).

The ultimate goal of science education is to develop scientific literacy, foster curiosity, encourage problem-solving, and prepare students for societal challenges like climate change and public health.

Role of the Teacher in Promoting Scientific Literacy

- In promoting scientific literacy, the teacher are the most important variable, therefore, it is not enough for the teacher to impart scientific and technological knowledge alone without letting the learners be aware of the inherent value of the knowledge to their own personal survival and the survival of the society in general.
- Teachers should be willing to help their students acquire and develop skills as well as provide scientific and technological knowledge. They should also be able to provide students with experience that permits them to use their skills and their knowledge of science and technology in making personal decisions.
- Teachers are also expected to make students develop positive attitude towards science and technology.

Best practices in science teaching and learning

Best practices in science teaching and learning are those ideal practices that ensure effective teaching and learning of science. These practices are expected to be imbibed by teachers, schools, government and other stakeholders.

1. Hands-On activities
2. Learning based on experience (constructivism)
3. Simulations
4. Use of simple, available materials (Improvisation)
5. Combining science with children's literature works
6. Standard based instructions
7. Setting high student expectations
8. Co-operative learning
9. Effective questioning
10. Authentic assessment work

Philosophical Views of Science

The philosophy of science is the branch of philosophy that studies the foundations, methods, and implications of science, asking what distinguishes science from non-science, how reliable scientific theories are, and what the ultimate goals of science should be. It explores questions about truth, evidence, reasoning, and the nature of scientific progress. Science has been interpreted through different philosophical lenses: Intuitivism emphasizes knowledge gained through direct perception or intuition, Falsificationism (Karl Popper) insists that scientific theories must be testable and refutable, and Paradigm theory (Thomas Kuhn) highlights how science progresses through shifts in dominant frameworks rather than steady accumulation of facts.

Intuitivism

- Rooted in the idea that some truths are grasped directly by intuition rather than empirical testing.
- Associated with early rationalist traditions (e.g., René Descartes) and later thinkers who argue that certain scientific insights are self-evident.
- Criticism: Intuition is subjective and unreliable compared to systematic observation.

- Intuitivism reminds us of the role of creativity and insight in discovery.

Falsificationism

- Developed by **Karl Popper** in *The Logic of Scientific Discovery* (1934).
- Core principle: A theory is scientific only if it can be **falsified** - meaning there is a possible observation or experiment that could prove it wrong.
- Example: Einstein's relativity predicted starlight bending during an eclipse. If observations had contradicted this, the theory would be refuted.
- Strength: Protects science from unfalsifiable claims, reminds us of the role of creativity and insight in discovery and sets a clear boundary between science and non-science.
- Limitation: Some scientific theories evolve by adjusting to anomalies rather than being discarded.

Paradigms

- Introduced by **Thomas Kuhn** in *The Structure of Scientific Revolutions* (1962).
- Science operates within a **paradigm** - shared framework of theories, methods, and assumptions.
- Progress occurs through **normal science** (working within the paradigm) until anomalies accumulate, leading to a **paradigm shift** (scientific revolution). Paradigm theory explains why scientific revolutions reshape entire fields rather than just adding new facts.
- Example: The shift from Newtonian mechanics to Einstein's relativity.
- Limitation: Suggests science is not purely cumulative but shaped by social and historical contexts.

Comparison Table

View	Core Idea	Strengths	Limitations
Intuitivism	Knowledge via intuition	Highlights role of insight	Subjective, lacks rigor
Falsificationism	Theories must be testable/refutable	Clear demarcation of science vs pseudo-science	Too rigid; ignores theory adaptation
Paradigms	Science shaped by dominant frameworks	Explains revolutions in science	Can underplay cumulative progress

Mode of scientific thinking

In **scientific inquiry**, reasoning is the backbone of how knowledge is built. The two main approaches are **inductive reasoning** and **deductive reasoning**, and they complement each other in the scientific process.

Inductive Reasoning

Inductive reasoning moves from specific observations to broader generalizations or theories. It involves the scientific process where Scientists collect data, notice patterns, and infer general principles. An example is observing that metals expand when heated → concluding that *all metals expand when heated*. It can generate new hypotheses and theories. But its

limitation is that conclusions are probable, not certain due to future observations that may contradict them.

Deductive Reasoning

Deductive reasoning starts with a general principle or theory and applies it to predict specific outcomes. It involves the process of Scientists testing hypotheses by deducing what should happen under certain conditions. **Example:** If *all metals expand when heated* → then heating iron should make it expand. Deductive reasoning provides logical certainty if premises are true. Its limitation is that it relies on the accuracy of the initial theory — if the premise is flawed, the conclusion will be too.

Both modes of reasoning work together in developing scientific knowledge. **Induction** helps scientists **form hypotheses** and theories. While **Deduction** allows them to **test predictions** derived from those theories. Example: Darwin used inductive reasoning to propose evolution by natural selection, while later experiments and genetic studies used deductive reasoning to test specific predictions of evolutionary theory.

Comparison Table

Reasoning Type	Direction	Example	Role in Science
Inductive	Specific → General	Observing many swans are white → inferring all swans are white	Builds theories
Deductive	General → Specific	All swans are white → predicting the next swan seen will be white	Tests theories

Scientific attitude and values

Scientific attitudes are the habits of mind that guide rational inquiry. Scientific attitudes focus on developing habits of thought and inquiry to shape how scientist thinks, and to acquire critical thinking and problem – solving skills. Examples include:

1. **Curiosity:** Drives questioning and exploration, motivating deeper investigation.
2. **Rationality:** Relies on logic and evidence rather than emotion or tradition.
3. **Open-mindedness:** Willingness to consider new ideas and revise beliefs when evidence demands.
4. **Skepticism:** Protects against misinformation by requiring proof/evidence before acceptance of claims, ideas, or knowledge.
5. **Intellectual honesty:** Reporting observations truthfully, even when results contradict expectations.
6. **Perseverance:** Continuing inquiry despite challenges or failures.

Scientific values

Scientific values emphasised the Ethical principles guiding the action of the scientist that shapes why we act responsibly: trustworthy science and social responsibility. Scientific values emphasize honesty, objectivity, perseverance, and a commitment to truth.

1. **Objectivity:** Judgments are based on facts, not personal bias.
2. **Integrity:** Upholding truth and transparency in research and communication.
3. **Respect for evidence:** Accepting conclusions only when supported by reliable data.
4. **Humility:** Recognizing the limits of knowledge and being open to correction.
5. **Commitment to progress:** Using science to improve human welfare and solve real-world problems.

Scientific attitude and values together shape how individuals think critically, solve problems, and engage responsibly with society. Encouraging these traits in students fosters lifelong learning and civic responsibility.

Ethical issues in Science education

Ethical issues in science revolve around ensuring research is conducted responsibly, transparently, and with respect for human rights, the environment, and society. The most pressing concerns today include data integrity, informed consent, genetic engineering, artificial intelligence, and environmental sustainability.

1. **Research integrity:** Fabrication, falsification, and plagiarism undermine trust in scientific findings. Therefore, Scientists must report results honestly, even if they contradict expectations.
2. **Artificial Intelligence (AI):** AI raises ethical concerns around bias, discrimination, privacy, and fairness. Since AI systems are used in decision making processes such as hiring, policing, and loan approvals, ensuring transparency, accountability, and fairness is critical.
3. **Genetic Engineering and CRISPR:** The advent of Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) allows precise DNA editing, offering potential cures for genetic diseases. However, editing human embryos sparks debates about “designer babies” and unknown long-term effects, challenging moral boundaries.
4. **Data Privacy and Surveillance:** Mass data collection by corporations and governments raises concerns about privacy and civil liberties. Tools like facial recognition and real-time tracking can infringe on freedoms, making balance between security and privacy essential.
5. **Informed consent:** Informed consent in research means participants voluntarily agree to take part in a study after being fully informed about its purpose, procedures, risks, and benefits. It is not just a signed form but an on-going process of communication, ensuring respect for autonomy and protection of participants.
6. **Balancing Innovation and Ethics:** Scientific progress must be weighed against ethical implications. The debate centres on whether benefits justify risks, and when ethical considerations should override technological advancement. This on-going debate is crucial for guiding responsible research and development.
7. **Environmental responsibility:** Climate change research, biodiversity studies, and sustainability initiatives must balance progress with ecological protection, to ensure minimal harm to the ecosystems and future generations.

Relationship between Science, Technology and Society

Science, technology, and society are interconnected in a continuous feedback loop where science generates knowledge, technology applies it, and society shapes both through needs, values, and ethical considerations.

Science as the Foundation

Science is the systematic pursuit of knowledge about the natural world through observation, experimentation, and analysis. It uncovers fundamental principles, from physics to biology, which form the basis for technological innovation. Scientific discoveries provide the theoretical framework that enables the development of new tools, processes, and solutions, allowing humans to understand and manipulate their environment in increasingly sophisticated ways

Technology as the Application

Technology translates scientific knowledge into practical applications. It encompasses everything from simple tools like the wheel to complex systems such as artificial intelligence. By applying scientific principles, technology enables society to solve problems, improve efficiency, and enhance quality of life. Technological advancements, in turn, often create new opportunities for scientific research, as improved instruments and methods allow scientists to explore previously inaccessible phenomena.

Society as the Guiding Force

Society influences both science and technology by determining priorities, allocating resources, and establishing ethical boundaries. **Social needs, cultural values, and public concerns shape the direction of research and technological development.** For example, societal focus on climate change has driven investment in renewable energy technologies, while ethical debates around genetic engineering have led to regulations guiding its application.

Science, technology, and society are deeply interdependent. Science provides knowledge, technology applies it, and society shapes both through values, ethics, and needs. The convergence of science, technology, and society has big impact on virtually every aspect of human life such as:

Healthcare: Scientific breakthroughs and technology have big impact in medicine, boosting life expectancy and quality of life: Eradication of diseases (e.g., polio), innovations like vaccines, antibiotics, imaging techniques, and advanced treatments.

Communication: The internet and mobile technologies have reshaped how people connect and access information. Instant global communication breaks down barriers, while social media fosters interconnectedness but also introduces challenges like misinformation and privacy concerns.

Transportation: Advances in automobiles, airplanes, high-speed trains, and autonomous vehicles have made travel faster, more efficient, and widely accessible. These innovations reduce distances, facilitating trade, tourism, and cultural exchange.

Agriculture: Science and technology have boosted agricultural productivity through genetically modified crops and precision farming. These methods improve yields, reduce waste, and strengthen food security for a growing global population.

EVOLUTION OF SCIENCE

The evolution of science has transitioned from basic **survival observations** into modern **quantum mechanics** and **artificial intelligence** through continuous global inquiry. This section will be discussed under the following era: Prehistoric Beginnings, Ancient Civilizations, Medieval Science, Islamic Golden Age, Scientific Revolution, and Modern Science and Technology (19th–21st Century).